DWSDiff: Dual-Window Spectral Diffusion for Hyperspectral Anomaly Detection

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***Abstract*— Anomaly detection (AD) has emerged as a critical area of research in hyperspectral imagery (HIS) processing, focusing on detecting sparse, small targets with spectral and spatial features deviating from the background without prior information. The approach of AD based on reconstruction differences is a leading method in deep learning (DL) for hyperspectral AD (HAD). A key challenge is the accurate estimation of complex backgrounds. The essence of this challenge lies in accurately reconstructing background regions while inferring the latent background of anomaly regions. In this article, we propose a novel method called dual-window spectral diffusion (DWSDiff) for HAD. To address the challenge of complex background estimation in HSIs, we developed a spectral diffusion model specifically tailored for HSI. This model achieves precise background estimation through an iterative spectral diffusion and reverse reconstruction process. We also introduced a dual-window strategy to mitigate the influence of anomaly extension areas within the neighborhood on background estimation. Moreover, the scarcity of paired labeled HSIs from the same scene, with and without anomalies, limits the model's ability to learn features between anomaly and background. To address the shortage, we devised an anomaly generation strategy based on the principal component analysis (PCA) and the linear spectral mixing model (LSMM). Building on these, we designed a training and inference framework that integrates spectral diffusion, reverse background reconstruction, and target detection. Experimental results on the Airport–Beach–Urban (ABU) hyperspectral datasets demonstrate that DWSDiff outperforms 20 state-of-the-art (SOTA) HAD methods across six different areas under the curve (AUC) metrics.**

***Index Terms*— Anomaly detection (AD), diffusion model, hyperspectral imagery(HSI).**

I. Introduction

HYPERSPECTRAL remote sensing captures spectral information across tens to hundreds of contiguous bands and integrates it with spatial data. This unique "image-spectrum integration" capability makes hyperspectral imaging indispensable across a wide range of fields, including agriculture [1], forestry [2], mineral exploration [3], environmental monitoring [4], and land cover surveys [5].

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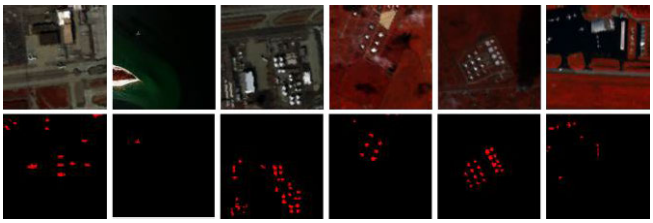


Fig. 1. Examples of various anomaly targets across different scenes.

Anomaly detection (AD), a crucial subfield of machine learning, has significant applications in computer vision. The primary objective of AD is to identify data that significantly deviate from expected or normal patterns. In the context of hyperspectral imaging, AD focuses on identifying observations that are inconsistent with the surrounding environment, making it an active research area in hyperspectral imagery (HSI) processing.

Hyperspectral AD (HAD) plays an essential role in applications such as environmental monitoring and national security, where detecting small, concealed, or unexpected targets is vital. HAD is a challenging yet critical task due to the lack of prior spectral and spatial information about targets and the complexity of background variability. Anomalies are typically sparse, small objects within a scene that exhibits spectral features markedly different from the surrounding background, such as airplanes, vehicles, and buildings (see Fig. 1).

Currently, HAD methods can be broadly categorized into three approaches: statistical modeling-based methods, representation learning-based methods, and deep learning (DL)-based methods. Among these, DL-based methods have received increasing attention in recent years due to their ability to learn complex patterns from data, achieving promising results in remote sensing tasks [6], [7], [8]. In the HAD field, reconstruction-based DL methods have demonstrated feasibility through extensive validation. However, these methods often emphasize the difficulty of reconstructing anomalies while neglecting the spectral differences between the target and the surrounding background, which is a crucial aspect of AD. Moreover, HAD methods often struggle to balance accurate background modeling and anomaly suppression, particularly in complex scenes.

Diffusion-based models have recently gained traction in the field of AD, primarily due to their ability to generate high-quality, realistic images that capture the complex distribution of backgrounds more effectively, thus improving the accuracy

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of AD. Although diffusion models have been applied in various HSI processing tasks [9], [10], [11], [12], [13], [14], [15], their application in HAD remains relatively underexplored. Existing diffusion models typically add spatial random noise to each spectral band of the original image without adequately considering spectral dimension information, leading to potential inaccuracies in spectral dimension reconstruction.

Given the data-driven nature of DL and the high cost of acquiring HSIs in real-world scenarios, the challenge is intensified by the lack of labeled data capturing both anomalous and normal scenes within the same context. In real remote sensing, anomalies have two key characteristics, diverse spatial morphology and spectral features distinct from the background, but aligned with the overall scene's spectral trends due to atmospheric effects. Current anomaly generation methods overlook these factors.

To address these challenges, we propose a novel method for HAD, termed dual-window spectral diffusion (DWSDiff) for HAD. Our contributions are given as follows.

1) We propose an anomaly spectrum generation method based on principal component analysis (PCA) decomposition and linear spectral mixing model (LSMM), addressing the issue of the lack of paired labeled data with and without anomalies, while also improving the realism over traditional anomaly generation methods.

2) We design a spectral diffusion model specifically for hyperspectral data, viewing the diffusion process of each pixel as a spectral mixture of the target and the surrounding background.

3) We introduce a DWSDiff background reconstruction method, effectively suppressing the target spectrum through iterative accumulation of spectra within a dual window that excludes the target.

The remainder of this article is organized as follows. Section II reviews current HAD methods. Section III outlines the preliminaries, including denoising diffusion models and the U-Net Model. Section IV details the proposed DWSDiff for HAD. Section V presents comprehensive experimental results comparing the proposed method with state-of-the-art (SOTA) methods. Finally, the conclusion is summarized in Section VI.

II. Related Work

Current HAD algorithms can be categorized into three main types: statistical modeling-based methods, representation learning-based methods, and DL-based methods. In recent years, extensive research has been conducted on these approaches.

A. Statistical Modeling-Based Methods

Statistical models assume that the background follows a specific statistical distribution, with anomalies that deviate from this distribution.

The Reed–Xiaoli anomaly detector (RX-AD) [16] is a milestone in this category, assuming a Gaussian background distribution and measuring anomalies using the Mahalanobis distance between the test pixel and the estimated distribution. Building on the classical RX detector, numerous extended methods have since been developed, such as local RX (LRX) [17], kernel RX (KRX) [18], weighted RX (W-RX) [19], 2S-GLRT [20], linear filter-based RX (LF-RX) [19], robust nonlinear AD (RNAD) [21], support vector data description (SVDD) [22], multiwindow adaptive detection (MWAD) [23], and AD based on guided filtering [24].

However, the statistical methods rely on assumptions about various distributions, which may not hold in complex backgrounds, leading to suboptimal detection results. Therefore, researchers have explored alternative theories for HAD, applying concepts such as local summation AD (LSAD), truncated statistic constant false alarm rate (TS-CFAR) [25], robust background regression-based score estimation (RBRSE) [26], fractional Fourier entropy (FrFE) [27], kernel isolation forest (KIFD) [28], isolation using nearest neighbor ensemble (iNNE) [29], and target detection through information entropy estimation based on point-set topology (TD-IEEPST) [30], all of which have achieved promising detection performance.

B. Representation Learning-Based Methods

Representation learning-based methods address the limitations of statistical approaches by eliminating the need for assumptions about the background's statistical distribution. These methods can be classified into two primary categories: data representation-based and data decomposition-based methods.

Data representation-based methods rely on the assumption that background pixels in hyperspectral data can be approximated by neighboring pixels or a background dictionary, while anomaly pixels cannot. The process typically involves constructing a background dictionary and sparse coefficient matrix and then using these to compute the reconstruction error of the test pixel to determine if it is an anomaly. Notable examples include sparse representation (SR) [31], [32] and collaborative representation (CR) [33]. SR-based methods highlight that normal/background samples can be expressed by a few atoms in an overcomplete dictionary, whereas anomaly samples cannot. On the other hand, CR-based methods focus on the collaborative relationships between dictionary atoms, analyzing whether each sample can be linearly represented by its neighbors. Both methods use reconstruction residuals to quantify the anomaly level of test pixels.

The second category, data decomposition-based methods, utilizes the fact that the background in hyperspectral data is typically composed of single components and occupies most of the pixels in the original data, while anomalous targets are generally rare. This allows for the decomposition of low-rank components from the original data to represent the background. Representative methods include low-rank representation (LRR) [34], [35], [36], robust PCA (RPCA) [37], [38], and Go-Dec-based methods [39], [40], [41]. In addition, HSIs can be treated as third-order tensors, enabling the application of tensor decomposition theory for AD. Examples of this approach include tensor decomposition AD (TDAD) [42], tensor PCA (TPCA) [43], and prior-based tensor approximation (PTA) [44].

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However, these representation-based methods depend on image-specific priors or dictionaries constructed for specific backgrounds. The construction of background dictionaries can be easily influenced by noise and anomalies, resulting in decreased detection effectiveness.

C. DL-Based Methods

In recent years, DL has shown exceptional performance in HAD, owing to its powerful capability for deep feature extraction.

Currently, reconstruction-based DL methods have become mainstream. These methods utilize the powerful data distribution learning abilities of DL to estimate the background distribution. Since sparsely distributed anomalies in the scene are difficult to reconstruct, AD is achieved based on reconstruction errors. Notable methods include those based on convolution neural networks (CNNs), autoencoders (AEs), generative adversarial networks (GANs), and transformer, such as GAED [45], RGAE [46], Auto-AD [47], FCAE-DCAC [48], MMBSN [49], LRR-Net [50], DFAN-HAD [51], DirectNet [52], HADGAN [53], CL-CaGAN [54], and TGFA-AD [55]. Some method combines LRR and DL [56], all of which have demonstrated excellent performance in AD.

Diffusion models have recently gained attention as a popular data generation approach. Compared to AE and GAN, diffusion models can achieve higher accuracy in background estimation, thereby improving detection performance. Although diffusion models have been widely used in various fields for AD [57], [58], [59], their application in HAD remains rare, with more research focused on hyperspectral classification [13], [15], image generation [14], super-resolution [9], [11], [12], and image fusion [10]. BSDM is one notable method that pioneers the use of diffusion models for HAD [60].

Most DL reconstruction models use the input HSI as the target of learning, meaning that the loss function is based on the error between the input containing anomalies and the reconstructed image. While these models aim the reconstruct the background, they may inadvertently overfit to anomalies, causing some anomalies to persist in the reconstructed HSI. This overlap makes it challenging to distinguish anomalies from the background, ultimately reducing model performance.

To address these issues, we generate anomalies consistent with the spectral features of the scene based on PCA and LSMM, ensuring the effectiveness of the learned features. To enhance the accuracy of spatial–spectral dimension estimation during reconstruction, we establish a spectral diffusion model, yielding high spatial and spectral precision in background estimation. Furthermore, to reduce the impact of the extension area of anomalies on background estimation, we design a dual-window diffusion process specifically for anomaly target detection.

III. Preliminaries

The denoising diffusion probabilistic model (DDPM) model [61] forms the foundation of the spectral diffusion approach used in this work, with both the endmember estimation network and the target detection network employing U-Net-like architectures. In this section, we briefly introduce the DDPM and the U-Net architecture.

A. Denoising Diffusion Models

Diffusion models, a type of generative model inspired by nonequilibrium thermodynamics, have gained significant attention. Among them, DDPMs [24] stand out as the most notable. DDPMs consist of a forward diffusion process, where noise is incrementally added to the real data, and a reverse diffusion process, where the noise is progressively removed to recover the original data. This approach allows for the generation of data with both interpretability and diversity. The forward diffusion process is defined as

$$x_t = x_0\sqrt{\bar{\alpha}_t} + \varepsilon_t\sqrt{1-\bar{\alpha}_t},\ \ \varepsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{1}$$

where x_0 represents the real data, which is transformed into noisy data x_t by progressively adding noise ε_t. The time step t is randomly sampled from $\{0, 1, \ldots, T\}$, and $\bar{\alpha}_t = \prod_{i=0}^{t} \alpha_i = \prod_{i=0}^{t} (1-\beta_i)$, where $\beta_i \in (0, 1)$ represents the preset noise variance. As t increases, the noise spreads throughout the original image, causing it to gradually approximate a standard Gaussian distribution.

The reverse diffusion process aims to denoise and gradually recover the original image from Gaussian noise, accomplished through a neural network. The network used for denoising typically follows a U-Net-like architecture. This network, denoted as $\hat{\varepsilon}_\theta(x_t, t)$, is trained to predict ε by minimizing the objective function

$$L = \mathbb{E}_{t, x_0, \varepsilon}\left\|\varepsilon_t - \hat{\varepsilon}_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{a}_t}\varepsilon, t\right)\right\|. \tag{2}$$

During backward processing, the trained network $\hat{\varepsilon}_\theta(x_t, t)$ reconstructs x_{t-1} from x_t as follows:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t, t)\right) + \tilde{\beta}_t z \tag{3}$$

where $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\tilde{\beta}_t = \beta_t(1-\bar{\alpha}_{t-1})/(1-\bar{\alpha}_t)$. Through the continuous iteration, the final reconstructed image x_0 is generated via the sequence $x_t \to x_{t-1} \to \cdots \to x_0$.

B. U-Net Model

The U-Net is a neural network architecture commonly used for image segmentation and generation tasks. Originally proposed by Ronneberger et al. [62], U-Net gained prominence for its superior performance in biomedical image segmentation. The core design of U-Net is based on an encoder–decoder structure, which facilitates the extraction of image features at multiple levels and their reconstruction layer by layer, making it particularly suitable for tasks requiring high-resolution outputs.

The U-Net architecture comprises a symmetric encoder and decoder, connected by skip connections. The encoder progressively downsamples the input image to extract high-level features, while the decoder upsamples these features to reconstruct the high-resolution output image.

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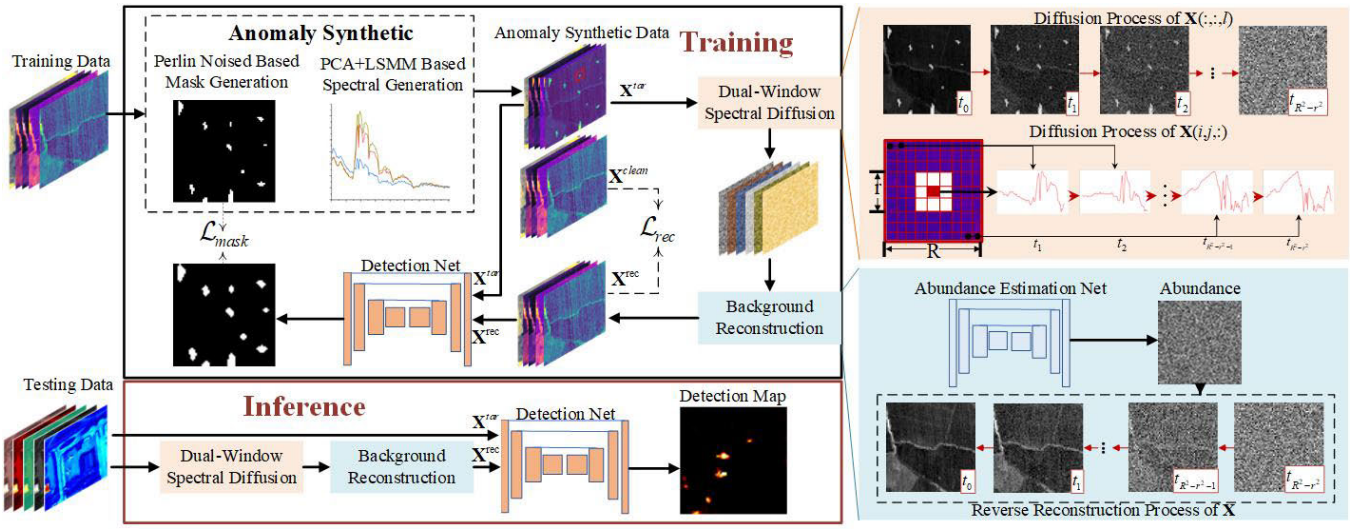


Fig. 2. Workflow of the DWSDiff-HAD method.

The encoder consists of several convolutional layers followed by pooling layers. Each convolution operation is followed by an ReLU activation function, and spatial downsampling is achieved through pooling layers, enabling the extraction of multiscale feature representations.

Each layer of the decoder consists of an upsampling layer followed by convolutional layers. The upsampling layer restores spatial resolution, and the convolutional layers refine the feature representations. The decoder also integrates multiscale information by merging features from the corresponding layers of the encoder through skip connections.

Skip connections directly transfer features from each layer of the encoder to the corresponding layer in the decoder, allowing the model to combine features from different levels, enhancing the detail and accuracy of the generated images.

IV. Proposed Method

In this section, we introduce the proposed DWSDiff-HAD. The method includes a dual-window diffusion process, a reverse background reconstruction process, and AD. The overall framework consists of two networks: the abundance estimation net and the detection net. The abundance estimation net estimates the abundance and then reconstructs the background $\mathbf{X}^{\text{rec}}$. The detection network predicts pixel-level anomalies by comparing $\mathbf{X}^{\text{rec}}$ with the input image $\mathbf{X}^{\text{tar}}$. Considering the lack of paired images with and without anomalies for training, we designed an anomaly generation strategy. The overall workflow is depicted in Fig. 2, which includes both the training and inference processes.

During the training process, the data $\mathbf{X}^{\text{tar}}$ containing synthetic anomalies are first generated from training data $\mathbf{X}^{\text{clean}}$. After diffusion and reconstruction, the background estimate $\mathbf{X}^{\text{rec}}$ is obtained. Both $\mathbf{X}^{\text{rec}}$ and $\mathbf{X}^{\text{tar}}$ are then fed into the detection net. The difference between the detection map and the anomaly mask, as well as the difference between $\mathbf{X}^{\text{rec}}$ and $\mathbf{X}^{\text{clean}}$, is used as loss functions to train both the abundance estimation net and the detection net.

In the inference process, the test image $\mathbf{X}^{\text{tar}}$ undergoes diffusion and reconstruction to produce the background $\mathbf{X}^{\text{rec}}$. This along with $\mathbf{X}^{\text{tar}}$ is then input into the trained detection net to generate the final detection results.

A. Anomaly Synthetic Strategy Based on PCA and LSMM

Due to the lack of labeled hyperspectral anomaly data, we employ an anomaly synthetic strategy to achieve end-to-end training. Real-world HSIs typically suffer from significant spectral distortion due to atmospheric absorption, atmospheric path radiation, solar illumination, and detector effects, leading to similar peaks and troughs in the entire scene. Current anomaly generation methods typically involve overlaying spectra from other objects onto target areas or directly replacing them with higher radiative energy. However, since real-world anomalies, such as airplanes, ships, cars, and houses, are often subpixel and not necessarily high-brightness targets, networks trained on data generated by these methods may have doubtful detection performance.

We propose an HAD data generation method based on PCA and a spectral mixing model to produce physically interpretable hyperspectral anomaly spectra. This method considers both the spatial morphology and spectral features of the target. The process is illustrated in Fig. 3.

1) Anomaly Spatial Morphology Generation: Considering the diverse shapes of anomaly targets, we utilize Perlin noise [63] to generate masks for the target regions. Perlin noise is a continuous and smooth random noise that can simulate various natural shapes. By adjusting the noise frequency and amplitude, we can generate target regions of different shapes and sizes.

2) Anomaly Spectral Generation: First, PCA is applied to the HSI for dimensionality reduction, yielding the principal component matrix $\mathbf{V}$ and the mean spectrum $\boldsymbol{\mu}$. Then, using a randomly generated weight vector $\mathbf{w}$, a new target spectrum $\mathbf{s}$ is reconstructed by principal components as follows:

$$\mathbf{s} = \boldsymbol{\mu} + \frac{1}{n_{\text{components}}}\mathbf{V}^{\text{T}}\mathbf{w}. \tag{4}$$

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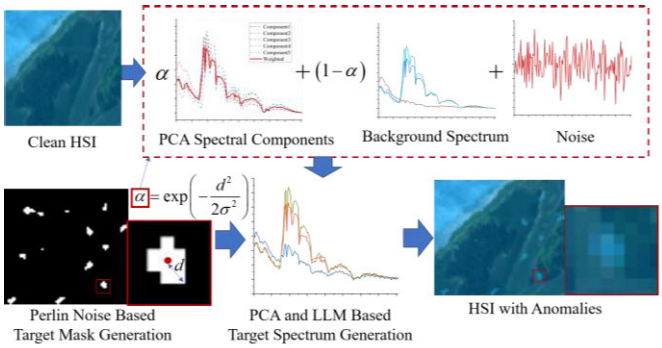


Fig. 3. Flowchart of the anomalies synthetic strategy.

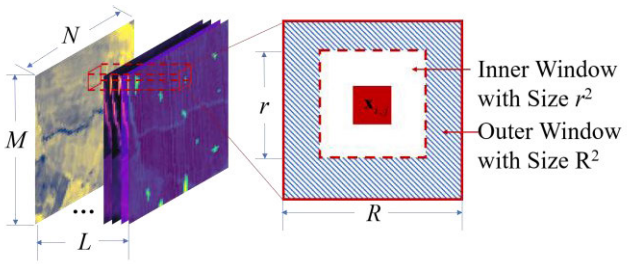


Fig. 4. Illustration of the dual window at location (i, j), where the shaded region represents the dual window with size R^2-r^2.

This method ensures that the generated target spectra are consistent with the overall spectral features of the original scene while exhibiting local spectral differences, effectively simulating the spectral properties of anomalies.

3) Gaussian-Weighted LSMM: Within the generated target region, spectral mixing is performed using a Gaussian distribution model. Assuming that the generated target region mask is $\mathbf{M}$, the target spectrum is $\mathbf{s}$, and the background spectrum (where $\mathbf{M} = 1$) is $\mathbf{X}_{\text{background}}$. The abundance α of each anomaly is calculated based on the distance d from the center to the edge of the target region

$$\alpha = \exp\left(-\frac{d^2}{2\sigma^2}\right) \tag{5}$$

where σ is the standard deviation of the Gaussian kernel. The abundance is highest at the center of the target region and gradually decreases toward the boundary. The final spectrum of the target region $\mathbf{X}_{\text{new}}$ is expressed as

$$\mathbf{X}_{\text{new}} = \alpha \mathbf{X}_{\text{background}} + (1-\alpha)\mathbf{s} + \mathbf{n} \tag{6}$$

where $\mathbf{n}$ is random noise added to increase data variability.

Going through the above process, the generated HSI not only simulates the spatial morphology of anomalies but also maintains consistency with the original scene's spectral features, providing physically interpretable and high-confidence AD training data.

B. Dual-Window Spectral Diffusion

The core challenge of reconstruction-based HAD is accurately estimating a clean background. Existing methods typically rely on the difficulty of reconstructing anomalies to estimate the background, but these approaches require strict training configurations. If the model's reconstruction ability is insufficient, background estimation accuracy decreases; if it is too strong, anomalies may also be reconstructed, reducing detection performance.

To address this issue, we propose a spectral diffusion model that progressively mixes target pixels with their surrounding spectral information. This causes anomalous pixels to gradually lose their distinctive features in the spectral dimension, eventually being replaced by the background spectra. During the reverse reconstruction process, the diffused data, when properly guided, can accurately restore the clean background. Given that anomalies usually have small geometric shapes, we introduce a dual-window strategy to minimize the influence of expanded anomaly regions during the diffusion process.

As shown in Fig. 4, for an HSI $\mathbf{X} \in \mathbb{R}^{M\times N\times L}$ and its (i, j)th pixel $\mathbf{x}_{i,j} \in \mathbb{R}^L$, we define a dual window centered on $\mathbf{x}_{i,j}$, and we define a dual window centered on $\mathbf{x}_{i,j}$ with an outer window size R and an inner window size r, where R and r are odd numbers. The dual window is obtained by subtracting the inner window from the outer window, leaving a hollow center. The outer window $\mathbf{W}^R_{i,j}$ and the inner window $\mathbf{W}^r_{i,j}$ are defined as

$$\mathbf{W}^R_{i,j} = \mathbf{X}\left[i-\frac{R-1}{2} : i+\frac{R+1}{2}+1,\ j-\frac{R-1}{2} : j+\frac{R-1}{2}+1, :\right] \tag{7}$$

$$\mathbf{W}^r_{i,j} = \mathbf{X}\left[i-\frac{r-1}{2} : i+\frac{r+1}{2}+1,\ j-\frac{r-1}{2} : j+\frac{r-1}{2}+1, :\right]. \tag{8}$$

We create a mask matrix $\mathbf{M} \in \mathbb{R}^{R\times R}$ defined as

$$\mathbf{M}_{u,v} = \begin{cases} 0, & \text{if } \left|u-\frac{R+1}{2}\right| < \frac{r}{2} \text{ and } \left|v-\frac{R+1}{2}\right| < \frac{r}{2} \\ 1, & \text{otherwise.} \end{cases} \tag{9}$$

The dual-window matrix was then obtained by elementwise multiplication of the outer window with the mask

$$\mathbf{W}^{\text{dual}}_{i,j} = \mathbf{W}^R_{i,j} \odot \mathbf{M} \tag{10}$$

where $\odot$ denotes the elementwise multiplication.

Next, a normal distribution abundance matrix $\mathbf{A} \in \mathbb{R}^{M\times N}$ is randomly generated, with each element $\mathbf{A}_{i,j} \sim \mathcal{N}(0, \sigma^2)$. The corresponding dual-window abundance matrix for the (i, j)th pixel $\mathbf{x}_{i,j}$ is

$$\mathbf{A}^{\text{dual}}_{i,j} = \mathbf{A}\left[i-\frac{R-1}{2} : i+\frac{R-1}{2}+1,\ j-\frac{R-1}{2} : j+\frac{R-1}{2}+1\right] \odot \mathbf{M}. \tag{11}$$

To simplify the notation in subsequent derivations, let the pixel at location (i, j) be denoted as $\mathbf{x}$ and the corresponding dual window as $\mathbf{W}$. The tth element in the dual-window

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abundance matrix $\mathbf{A}_{i,j}^{\text{dual}}$ is denoted as $a_t \sim \mathcal{N}(0, \sigma^2), t = 1, 2, \ldots, R^2 - r^2$.

Assuming that the original pixel before diffusion is $\mathbf{x}_0$, after adding the first element from the window

$$\mathbf{x}_1 = \mathbf{x}_0 + a_1 \mathbf{w}_1. \tag{12}$$

After adding the second element

$$\mathbf{x}_2 = \mathbf{x}_1 + a_2 \mathbf{w}_2 = \mathbf{x}_0 + a_1 \mathbf{w}_1 + a_2 \mathbf{w}_2. \tag{13}$$

Continuing to add until the tth element

$$\begin{aligned} \mathbf{x}_t &= \mathbf{x}_{t-1} + a_t \mathbf{w}_t \\ &= \mathbf{x}_0 + a_1 \mathbf{w}_1 + a_2 \mathbf{w}_2 + \ldots + a_t \mathbf{w}_t. \end{aligned} \tag{14}$$

Since $a \sim \mathcal{N}(0, \sigma^2)$, we have

$$\mathbf{x}_t \sim \mathcal{N}\left(\mathbf{x}_0, \sigma^2 \sum_{i=1}^{t} \mathbf{w}_t^2\right). \tag{15}$$

Thus,

$$\mathbf{x}_t = \mathbf{x}_0 + a\sqrt{\sum_{i=1}^{t} \mathbf{w}_t^2} \text{ where } a \sim \mathcal{N}\left(0, \sigma^2\right). \tag{16}$$

Assuming that $\boldsymbol{\beta}_i = \mathbf{w}_i^2$ and $\bar{\boldsymbol{\beta}}_i = \boldsymbol{\beta}_1 + \boldsymbol{\beta}_2 + \ldots + \boldsymbol{\beta}_i = \mathbf{w}_1^2 + \mathbf{w}_2^2 + \ldots + \mathbf{w}_i^2$, we have

$$\mathbf{x}_t = \mathbf{x}_0 + \sqrt{\bar{\boldsymbol{\beta}}_t} a. \tag{17}$$

During the diffusion process, the following distribution exists:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) \sim \mathcal{N}\left(\mathbf{x}_{t-1}, \sigma^2 \boldsymbol{\beta}_t\right) \tag{18}$$

$$q(\mathbf{x}_{t-1} | \mathbf{x}_0) \sim \mathcal{N}\left(\mathbf{x}_0, \sigma^2 \bar{\boldsymbol{\beta}}_{t-1}\right) \tag{19}$$

$$q(\mathbf{x}_t | \mathbf{x}_0) \sim \mathcal{N}\left(\mathbf{x}_0, \sigma^2 \bar{\boldsymbol{\beta}}_t\right). \tag{20}$$

From (20), it is clear that each pixel is a random variable with a mean of $\mathbf{x}_0$ and a variance of $\sigma^2 \bar{\boldsymbol{\beta}}_t$.

C. Reverse Reconstruction

The reverse reconstruction process involves iteratively estimating $\mathbf{x}_0$ based on the diffused $\mathbf{x}_t$ via the sequence $\mathbf{x}_t \to \mathbf{x}_{t-1} \to \cdots \to \mathbf{x}_0$. The first step is to estimate $\mathbf{x}_{t-1}$ based on $\mathbf{x}_0$.

To simplify the notation, all vector operations in the equations are elementwise. The true posterior distribution can be derived using Bayes' theorem, and the denoising process from $\mathbf{x}_0$ and $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$ can be expressed as

$$\begin{aligned} q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) &= \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)} \\ &\sim \mathcal{N}\left(\mathbf{x}_{t-1}; \frac{\bar{\boldsymbol{\beta}}_{t-1} \mathbf{x}_t + \boldsymbol{\beta}_t \mathbf{x}_0}{\bar{\boldsymbol{\beta}}_t}, \frac{\sigma^2 \boldsymbol{\beta}_t \bar{\boldsymbol{\beta}}_{t-1}}{\boldsymbol{\beta}_t}\right). \end{aligned} \tag{21}$$

Given the length of the derivation, the detailed process is provided in the Appendix.

In this article, we estimate the mean of the distribution $\mathbf{x}_{t-1}$. According to (21), $\mathbf{x}_{t-1}$ can be expressed as

$$\begin{aligned} \mathbf{x}_{t-1} &= \frac{\bar{\boldsymbol{\beta}}_{t-1} \mathbf{x}_t + \boldsymbol{\beta}_t \mathbf{x}_0}{\bar{\boldsymbol{\beta}}_t} \\ &= \frac{\bar{\boldsymbol{\beta}}_{t-1} \mathbf{x}_t + \boldsymbol{\beta}_t \left(\mathbf{x}_t - \sqrt{\bar{\boldsymbol{\beta}}_t} a_t\right)}{\bar{\boldsymbol{\beta}}_t} \\ &= \mathbf{x}_t - \frac{\boldsymbol{\beta}_t}{\sqrt{\bar{\boldsymbol{\beta}}_t}} a_t. \end{aligned} \tag{22}$$

For image generation tasks, the problem of estimating the HSI can be transformed into the task of estimating the abundance matrix. Therefore, the loss function can be defined as

$$\mathcal{L}_{\text{abundance}} = \mathbb{E}_{t, \mathbf{x}_0, a} \left\| a_t - \widehat{\mathcal{U}}_a \left(\mathbf{x}_0 + \sqrt{\bar{\boldsymbol{\beta}}_t} a_t, t\right) \right\|^2 \tag{23}$$

where $\mathcal{U}_a$ represents the trained neural network used to estimate the abundance matrix.

During the reverse reconstruction process, according to (20), the diffused image $\mathbf{x}_t$ is gradually restored to $\mathbf{x}_0$ by

$$\mathbf{x}_{t-1} = \mathbf{x}_t - \frac{\boldsymbol{\beta}_t}{\sqrt{\bar{\boldsymbol{\beta}}_t}} \mathcal{U}_a(\mathbf{x}_t, t). \tag{24}$$

D. Loss Function

In AD tasks, the accuracy of background estimation is critical. Therefore, the reconstruction loss is employed as the loss function

$$\mathcal{L}_{\text{rec}} = \frac{1}{MN} \sum_{i=1}^{MN} \left(\mathbf{X}_i^{\text{clean}} - \mathbf{X}_i^{\text{rec}}\right)^2 \tag{25}$$

where $\mathbf{X}_i^{\text{rec}}$ represents the reconstruction result of the ith pixel of the HSI $\mathbf{X}^{\text{tar}}$, which includes synthetic anomalies. This pixel is first diffused to $\mathbf{x}_t$ using (16) and then sampled back to $\mathbf{x}_0$ in the reverse process via (24).

For the detection component, a U-Net architecture is employed

$$\hat{\mathbf{M}} = \mathcal{U}_M\left(\mathbf{X}^{\text{tar}}, \mathbf{X}^{\text{rec}}\right). \tag{26}$$

Thus, the detection loss is defined as

$$\mathcal{L}_{\text{mask}} = \text{Smooth}_{L_1}\left(\mathbf{M}, \hat{\mathbf{M}}\right) + \gamma L_{\text{focal}}\left(\mathbf{M}, \hat{\mathbf{M}}\right) \tag{27}$$

where $\mathbf{M}$ is the ground-truth mask and γ is the weight balancing these two loss functions.

The combined loss function is

$$\mathcal{L}_{\text{total}} = k_{\text{rec}} \mathcal{L}_{\text{rec}} + k_{\text{mask}} \mathcal{L}_{\text{mask}} \tag{28}$$

where k_1 and k_2 are the weights for adjusting the importance of reconstruction and detection losses, respectively.

E. Training and Inference

The training and inference procedures for the DWSDiff model are outlined in the following two algorithms.

Algorithm 1 illustrates the training process, where the abundance estimation network $\mathcal{U}_a$ and the detection network $\mathcal{U}_M$ are developed. The model is trained by clean and anomaly

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Algorithm 1 Training

Input: n pairs of clean HSIs and synthetic anomaly HSIs: $\{(\mathbf{X}^{\text{clean}_1}, \mathbf{X}^{\text{tar}_1}), (\mathbf{X}^{\text{clean}_2}, \mathbf{X}^{\text{tar}_2}), \ldots, (\mathbf{X}^{\text{clean}_n}, \mathbf{X}^{\text{tar}_n})\}$, each HSI $\mathbf{X} \in \mathbb{R}^{M\times N\times L}$,width of inner window: r, width of outer window: R, variance of abundance matrix σ^2, number of epochs: E, batch size: B.

Output: Trained abundance estimation network $\mathcal{U}_a$, trained detection network $\mathcal{U}_M$

for epoch $= 1, \ldots, E$ **do**
  Shuffle $\{(\mathbf{X}^{\text{clean}}, \mathbf{X}^{\text{tar}})\}$ into batches of size B
  for each batch $\{\mathbf{X}^{\text{clean}_b}, \mathbf{X}^{\text{tar}_b}\}$ *in batches* **do**
    for each $(\mathbf{X}^{\text{clean}}, \mathbf{X}^{\text{tar}})$ *in* $\{\mathbf{X}^{\text{clean}_b}, \mathbf{X}^{\text{tar}_b}\}$ **do**
      for each pixel $i = 1, \ldots, M \times N$ **do**
        Construct $\mathbf{x}$, $\boldsymbol{\beta}_{1,\ldots,R^2-r^2}$, $\bar{\boldsymbol{\beta}}_{1,\ldots,R^2-r^2}$ at pixel $\mathbf{X}_i^{\text{tar}}$ and its corresponding dual window $\mathbf{W}$;
        Generate $\mathbf{A}^{double} \in \mathbb{R}^{R^2-r^2}$ with every element $a \sim \mathcal{N}(0, \sigma^2)$;
        Generate $t \sim Uniform(\{1, \ldots, R^2 - r^2\})$;
        Calculate $\mathbf{x}_t$ by (16);
        Calculate $\hat{\mathbf{x}}_0$ (also $\mathbf{X}_i^{\text{rec}}$) by (24);
      end for
      Obtain reconstruction data $\mathbf{X}^{\text{rec}}$ for the image
      Calculate the detection result $\hat{\mathbf{M}}$ by (25);
    end for
    Update $\mathcal{U}_a$ and $\mathcal{U}_M$ by minimizing loss function (28)
  end for
end for

Algorithm 2 Inference

Input: Observed HSI $\mathbf{X} \in \mathbb{R}^{M\times N\times L}$,width of inner window R, width of outer window r, variance of abundance matrix σ^2.

Output: Detection result $\hat{\mathbf{M}}$

for $i = 1, \ldots, M \times N$ **do**
  Construct $\mathbf{x}$, $\boldsymbol{\beta}_{1,\ldots,R^2-r^2}$, $\bar{\boldsymbol{\beta}}_{1,\ldots,R^2-r^2}$ at pixel $\mathbf{X}_i$ and its corresponding dual window $\mathbf{W}$;
  Generate $\mathbf{A}^{double} \in \mathbb{R}^{R^2-r^2}$ with every element $a \sim \mathcal{N}(0, \sigma^2)$;
  Generate $t = R^2 - r^2$;
  Calculate $\mathbf{x}_t$ by (16);
  Calculate $\hat{x}_0$(also $\mathbf{X}_i^{\text{rec}}$) by (24);
end for
Obtain the reconstruction data $\mathbf{X}_{\text{rec}}$;
Calculate the detection result $\hat{\mathbf{M}} = \mathcal{U}_M(\mathbf{X}, \mathbf{X}_{rec})$.

synthetic HSIs to learn how to accurately reconstruct the background and detect anomalies.

Once the model is trained, Algorithm 2 demonstrates how it is applied to new HSIs for AD. The inference process involves using the trained model to reconstruct the background and generate a detection map for the input data.

Since the diffusion and reconstruction processes are performed in a pixelwise manner, the input to the diffusion model at each iteration is the local neighborhood around the corresponding pixel, with dimensions $R \times R \times L$.

V. Experiments and Analysis

A. Experimental Settings

1) Datasets: We evaluated the detection capability of our proposed method using the Airport–Beach–Urban (ABU) HSI datasets, which are available online. The ABU dataset includes 13 different complex scenes, such as airports, beaches, and urban areas, encompassing a variety of targets such as ships, minerals, vehicles, and airplanes. These datasets were manually annotated based on images obtained from the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), with the exception of the Beach-4 dataset, which is derived from images captured by the Reflective Optical System Imaging Spectrometer (ROSIS-03). The information on these images is detailed in Table I.

TABLE I
Data Information of the ABU Testing Sets

Data name	Spatial resolution	Pixel number	Band number
Airport1	7.1m	100×100	205
Airport2	7.1m	100×100	205
Airport3	7.1m	100×100	205
Airport4	3.4m	100×100	191
Beach1	17.2m	150×150	188
Beach2	7.5m	100×100	193
Beach3	4.4m	100×100	188
Beach4	1.3m	150×150	102
Urban1	17.2m	100×100	204
Urban2	17.2m	100×100	207
Urban3	3.5m	100×100	191
Urban4	7.1m	100×100	205
Urban5	7.1m	100×100	205

TABLE II
Data Information of the AVIRIS Training Sets

Flight ID	Location	Date	Resolution(m)	Training Images
f111109t01	Los Angeles	2011/09/11	7.1	50
f110523t01	New Mexico	2011/05/23	16	50
f100513t01	Florida	2010/05/13	17.2	50
f080611t01	Moffett Field	2008/06/11	17	50
Total				200

The training data were sourced from publicly available AVIRIS datasets. Detailed information is provided in Table II. These HSIs cover various types of scenes, including urban areas, grasslands, forests, and coastal regions, with differing resolutions and imaging times. The data consist of 224 bands, and after removing all water vapor absorption bands (108–110 and 155–168), 207 bands remain. For each image, 50 subimages were randomly selected from typical regions.

2) Comparison Methods: We selected 20 different classic and SOTA HAD algorithms for comparison. These methods

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TABLE III
LIST OF COMPARISON METHODS

Method type	Method Name
Statistical Modeling-Based Methods	GRX[16], LRX[17],WRX[19], 2S-GLRT[20], FrFE[27], KIFD[28]
Representation Learning-Based Methods	LRASR[31], CRD[33], LSDM-MoG[35], RPCA-RX[34], GTVLRR[36], Go-Dec-RX[41], PTA[44]
Deep Learning-Based Methods	GAED, RGAE, Auto-AD[47], DFAN-HAD[51], DirectNet[52], BSDM[60], TGFA[55]

are categorized into three groups: statistical modeling-, representation learning-, and DL-based methods. The specific methods are listed in Table III.

Among the statistical modeling methods, GRX, LRX, and 2S-GLRT are well-known approaches that perform AD on HSIs under the assumption of a Gaussian background. FrFE employs the FrFE to detect anomalies. KIFD applies a novel approach using kernel isolation forest.

Sparse learning methods are categorized into data representation and data decomposition. For data representation methods, we considered two approaches: LRASR, which utilizes SR, and CRD, which employs CR. The comparison also covers four mainstream directions of data decomposition: LRR (LSDM-MoG and GTVLRR), RPCA-RX, Go-Dec-based (Go-Dec-RX), and tensor decomposition-based (PTA) methods.

In the category of DL-based methods, we included GAED, RGAE, Auto-AD, DFAN-HAD, DirectNet, BSDM, and TGFA. These methods all measure the anomaly score for each pixel based on the difference between the original and reconstructed images, which aligns with the motivation behind our proposed method. DirectNet utilizes a dual-window approach. Notably, BSDM also utilizes diffusion models for image reconstruction and then implements AD.

All methods were implemented using their official code, with default algorithm parameters. Data preprocessing was conducted according to the official guidelines. The majority of the methods employ max–min normalization during data preprocessing, adjusting each test HSI to fit within the [1, 0] range. However, some methods utilize distinct normalization strategies. For example, 2S-GLRT and PTA perform max–min normalization on each HSI band separately, whereas FrFE normalizes the entire test HSI by scaling all values relative to the maximum value.

3) Evaluation Metrics: To evaluate the performance of different HAD algorithms, we used eight area under the curve (AUC) metrics based on 3-D receiver operating characteristics (3DROCs) [64] and separability maps.

The 3-D ROC evaluation framework, represented by (PD, PF, τ), expands the traditional 2-D ROC model (PD, PF) by incorporating a threshold τ, thereby introducing an additional dimension. This enhancement allows for the creation of three distinct 2-D ROC curves: (PD, PF), (PD, τ), and (PF, τ). The corresponding AUC from these 2-D ROC curves, known as $\mathrm{AUC}_{(\mathrm{D},\mathrm{F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, and $\mathrm{AUC}_{(\mathrm{F},\tau)}$, can be used to evaluate detection performance. In addition, five combinations of these AUCs can be quantitatively analyzed to evaluate different aspects of detection capabilities.

Higher values of $\mathrm{AUC}_{(\mathrm{D},\mathrm{F})}$ and $\mathrm{AUC}_{(\mathrm{D},\tau)}$ indicate superior detection performance. Therefore, these values are combined to define $\mathrm{AUC}_{\mathrm{TD}}$, which reflects the detector's ability to accurately identify targets

$$0 \leq \mathrm{AUC}_{\mathrm{TD}} = \mathrm{AUC}_{(\mathrm{D},\mathrm{F})} + \mathrm{AUC}_{(\mathrm{D},\tau)} \leq 2. \tag{29}$$

$\mathrm{AUC}_{\mathrm{BS}}$ represents the detector's capability to minimize false positives due to background interference

$$-1 \leq \mathrm{AUC}_{\mathrm{BS}} = \mathrm{AUC}_{(\mathrm{D},\mathrm{F})} - \mathrm{AUC}_{(\mathrm{F},\tau)} \leq 1. \tag{30}$$

Overall detection probability (ODP) evaluates the combined effectiveness of target detectability and background suppressibility. It is calculated using $\mathrm{AUC}_{\mathrm{ODP}}$, which integrates both metrics to evaluate overall detection capability

$$-1 \leq \mathrm{AUC}_{\mathrm{ODP}} = \mathrm{AUC}_{(\mathrm{D},\mathrm{F})} + \mathrm{AUC}_{(\mathrm{D},\tau)} - \mathrm{AUC}_{(\mathrm{F},\tau)} \leq 2. \tag{31}$$

$\mathrm{AUC}_{\mathrm{TD-BS}}$ specifically evaluates the detector's performance in identifying targets within background noise by accounting for both detectability and background suppression

$$-1 \leq \mathrm{AUC}_{\mathrm{TD\text{-}BS}} = \mathrm{AUC}_{(\mathrm{D},\tau)} - \mathrm{AUC}_{(\mathrm{F},\tau)} \leq 1. \tag{32}$$

Assuming that PF is predominantly caused by noise, this metric is analogous to the signal-to-noise ratio (SNR). $\mathrm{AUC}_{\mathrm{SNPR}}$ reflects the detector's ability to distinguish signals from noise

$$0 \leq \mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}_{(\mathrm{D},\tau)}}{\mathrm{AUC}_{(\mathrm{F},\tau)}}. \tag{33}$$

Separability maps are introduced as a visual tool used to assess the effectiveness of AD methods. These maps depict the distribution of response values for both anomalies and backgrounds across different detection methods. By showing the statistical range (from the 10th to 90th percentiles) and the median of these values, separability maps allow for a clear comparison of how well each method separates anomalies from the background. Ideally, effective AD is indicated by narrow and low-lying background response distributions, with a distinct separation from the higher anomaly response distributions.

4) Implementation Details: The abundance estimation network, following diffusion, is based on a U-Net architecture enhanced with techniques from PixelCNN, ResNet, and transformers. These enhancements enable the network to capture more complex features and improve estimation accuracy compared to a standard U-Net. Similar to DDPM, a TimeEmbedding module encodes the timestep into the network during each training iteration.

Detection net also employs a U-Net architecture, consisting of an encoder, a decoder, and skip connections. The input to this model is a concatenation of the original image and the estimated background image along the channel dimension. The network predicts anomalies by analyzing the similarities and inconsistencies between these inputs. Key parameters include

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an outer window width $R = 13$, an inner window width $r = 5$, and the diffusion variance of the abundance matrix set to 0.5.

In our experiments, all data are standardized to a size of $100 \times 100 \times 200$. For datasets with more than 200 spectral bands, the first 200 bands are selected; for those with fewer than 200 bands, the last few bands are duplicated until the total reaches 200. Given that the U-Net architecture involves multiple downsampling and upsampling operations, the input image's spatial dimensions must be powers of two to ensure stable output sizes and proper execution of skip connections. Therefore, when using DWSDiff for detection, images are downsampled to $64 \times 64 \times 200$ to fit the UNet's requirements, and the detection map is then upsampled back to 100×100.

The training data are sourced from publicly available AVIRIS HSIs, with diverse training samples generated in real time using our proposed anomaly synthetic strategy. During training, each clean image was paired with a corresponding synthetic anomaly image generated using our proposed anomaly generation strategy. This process was applied to all 200 images to train the abundance estimation network and the detection network. The trained network is then tested on the ABU dataset.

We use Perlin noise the generate an anomaly mask. The distribution and sparsity of the simulated anomalies are controlled using Perlin resolution and segmentation threshold. Perlin resolution determines the spatial frequency and variability of the noise, which influences the size and shape of the anomaly regions. Segmentation threshold filters the Perlin noise to generate binary masks, where a higher threshold results in smaller and sparser anomalies.

Fig. 5 demonstrates the effect of varying Perlin resolution (3, 4, and 5) and threshold values (0.6, 0.7, and 0.8) on the generated anomaly masks. Based on observations from the ABU dataset, where anomalies are typically smaller than 5×5, we randomly select Perlin resolutions of 3 or 4 and thresholds in the range of [0.6, 0.8] during training. This ensures that the generated anomalies are diverse in shape, size, and sparsity while remaining consistent with real-world anomaly characteristics.

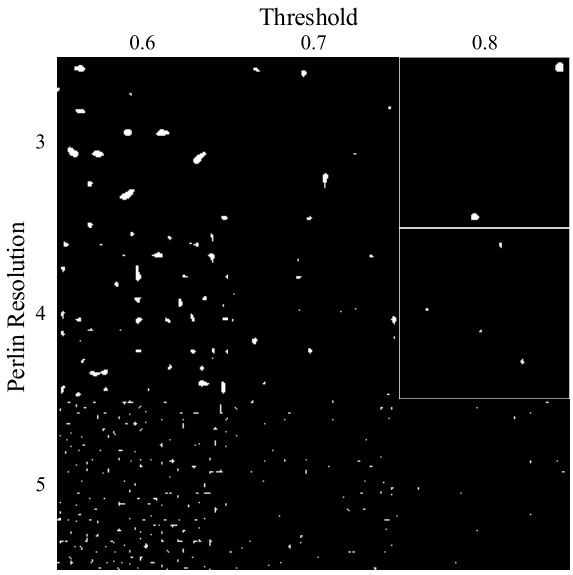


Fig. 5. Anomaly masks generated with varying Perlin resolutions (3, 4, and 5) and segmentation thresholds (0.6, 0.7, and 0.8). Higher Perlin resolutions and thresholds produce smaller and sparser anomaly regions.

B. Experiment Result

1) Comparisons of Detection Maps: Fig. 6 presents the detection maps of 21 different HAD methods on the Airport-1, Airport-3, Urban-1, Urban-4, Beach-1, and Beach-3 datasets. In each figure, the first and second images show the HSI in false color and the corresponding ground truth, respectively. These figures cover detection results in both complex and simple backgrounds, including scenarios with single and multiple targets.

From the figures, it is evident that different methods exhibit distinct characteristics in AD. Methods such as GRX, 2S-GLRT, FrFE, RPCA-RX, GoDec-RX, and Auto-AD tend to focus on background suppression, resulting in cleaner background areas with lower anomaly scores in their detection maps. On the other hand, other comparison methods emphasize enhancing the anomaly scores of the targets.

All comparison methods improve target–background contrast to some extent, but they also tend to misclassify background areas as targets. In contrast, the proposed DWSDiff method effectively suppresses the background while significantly enhancing anomaly contrast. This advantage is particularly evident in complex scenes such as Airport-3 and Urban-1.

2) Comparisons of AUCs of 3-D ROC: Table IV presents the average results of eight mAUC metrics across all test data for different AD algorithms. An upward arrow indicates that a higher value corresponds to better detection performance, while a downward arrow indicates the opposite. The proposed DWSDiff method outperforms the other 20 methods in six key metrics: $\text{mAUC}_{(\text{D},\text{F})}$, $\text{mAUC}_{(\text{D},\tau)}$, mAUC_{TD}, mAUC_{BS}, mAUC_{ODP}, and $\text{mAUC}_{\text{TD}-\text{BS}}$. These results indicate that DWSDiff demonstrates superior background suppression and target detectability compared to existing algorithms. Table V shows the $\text{AUC}_{(\text{D},\text{F})}$ for each method on each test dataset, along with the average $\text{AUC}_{(\text{D},\text{F})}$ across all datasets for each method. This table supports a similar conclusion as the $\text{mAUC}_{(\text{D},\text{F})}$ results in Table IV. Some algorithms exhibit significant variations in detection performance across different datasets. For instance, the TGFA algorithm performs exceptionally well in Urban-2, surpassing other methods, but falls short in Beach-1. In contrast, DWSDiff-HAD consistently delivers stable and outstanding detection performance across all data, demonstrating its robustness and adaptability.

We present the 3DROC and (PD, PF) 2DROC curves for 21 methods across three scenarios in Fig. 7. Although the proposed method does not achieve the absolute best performance on every data, it consistently ranks among the top across all test data. The overall performance of the DWSDiff is superior to other methods.

3) Comparisons of Separability Maps: Separability maps visually demonstrate the effectiveness of different methods in distinguishing between anomalies and background. The goal is to observe narrower green boxes located at the bottom, with

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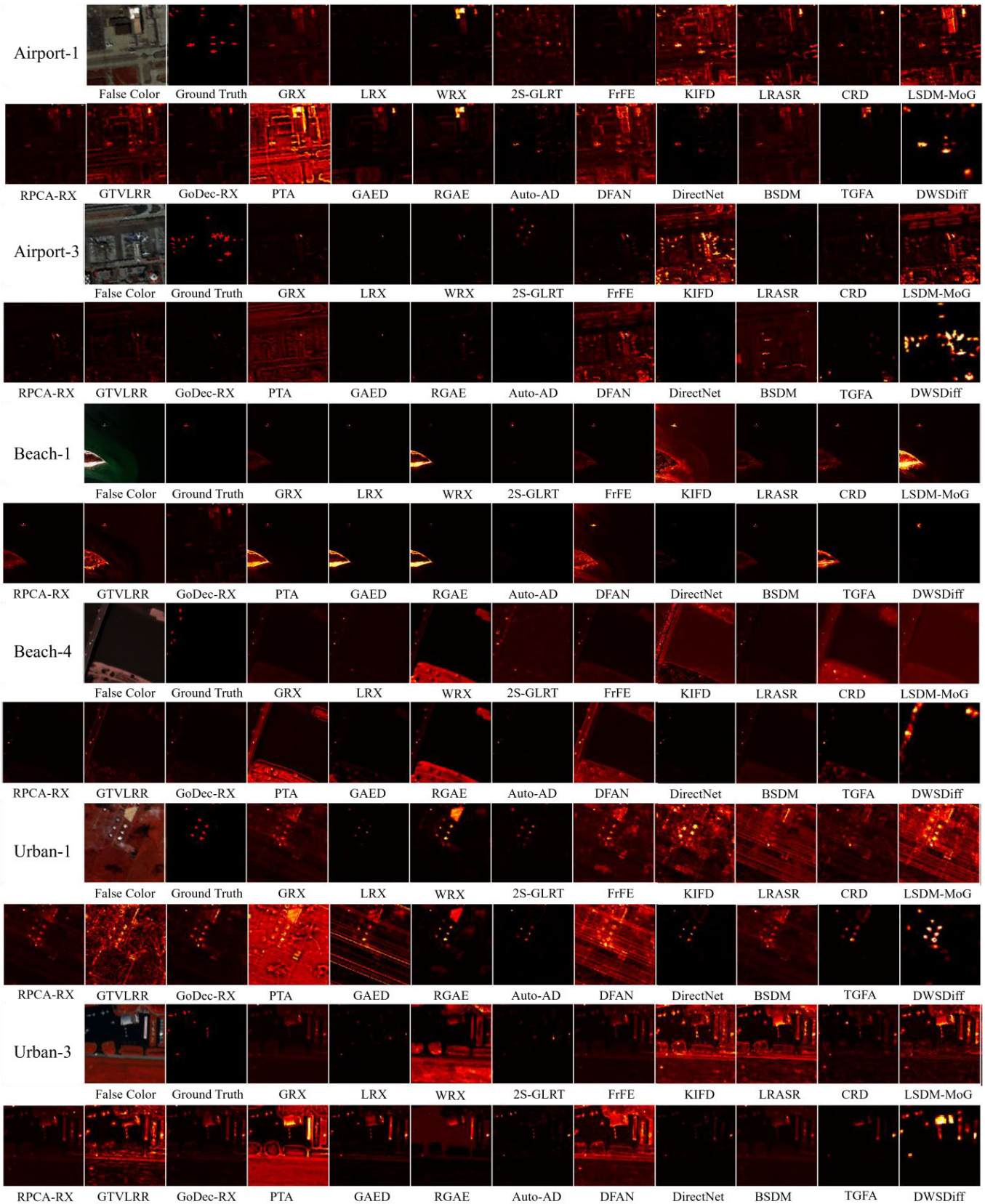


Fig. 6. Detection maps of different HAD methods applied to different data.

a clear separation from the red boxes, indicating strong AD and background suppression performance.

Fig. 8 presents the separability maps for three distinct scenes. From these results, it is evident that the DWSDiff

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TABLE IV

EIGHT MAUCS RESULTS FOR 21 HAD ALGORITHMS ACROSS THE ABU DATASET. THE TOP THREE PERFORMING METHODS ARE HIGHLIGHTED IN BOLD RED, GREEN, AND BLUE IN EACH COLUMN

.Methods	$mAUC_{(D,F)}\uparrow$	$mAUC_{(D,\tau)}\uparrow$	$mAUC_{(F,\tau)}\downarrow$	$mAUC_{TD}\uparrow$	$mAUC_{BS}\uparrow$	$mAUC_{ODP}\uparrow$	$mAUC_{TD\text{-}BS}\uparrow$	$mAUC_{SNPR}\uparrow$
GRX	0.94487	0.17341	0.03054	1.11828	0.91433	1.08774	0.14287	8.40640
LRX	0.93287	0.09570	**0.00451**	1.02856	0.92836	1.02406	0.09119	**175.63179**
WRX	0.84185	0.10921	0.03910	0.95106	0.80275	0.91196	0.07011	13.00492
2S-GLRT	0.91897	0.15553	0.01654	1.07450	0.90242	1.05795	0.13899	30.64549
FrFE	0.95846	0.18929	0.04085	1.14775	0.91760	1.10689	0.14843	9.78278
KIFD	0.95057	**0.52659**	0.10512	**1.47716**	0.84544	**1.37203**	**0.42147**	5.21445
LRASR	0.93206	0.19764	0.04048	1.12970	0.89158	1.08922	0.15716	7.49794
CRD	0.92414	0.17330	0.02757	1.09744	0.89657	1.06987	0.14573	9.33249
LSDM-MoG	0.94021	0.39296	0.09200	1.33317	0.84821	1.24118	0.30097	4.66734
RPCA-RX	0.94513	0.17530	0.02147	1.12043	0.92366	1.09895	0.15383	11.78557
GTVLRR	0.94528	0.32978	0.06239	1.27506	0.88288	1.21267	0.26739	6.13311
GoDec-RX	**0.96094**	0.14857	0.01572	1.10951	**0.94523**	1.09380	0.13286	16.14349
PTA	0.85609	0.33713	0.12949	1.19322	0.72660	1.06373	0.20764	3.27390
GAED	0.90756	0.15556	0.02072	1.06312	0.88684	1.04240	0.13484	13.79983
RGAE	0.87615	0.14519	0.02401	1.02135	0.85214	0.99734	0.12119	14.38906
Auto-AD	0.95028	0.15303	**0.00627**	1.10331	0.94400	1.09704	0.14676	**52.05284**
DFAN	**0.96552**	**0.45211**	0.09603	**1.41762**	0.86949	**1.32160**	**0.35608**	5.27753
DirectNet	0.95193	0.12799	**0.00484**	1.07993	**0.94709**	1.07508	0.12315	49.52139
BSDM	0.94486	0.17342	0.03055	1.11828	0.91431	1.08773	0.14287	8.40576
TGFA	0.92655	0.14559	0.00629	1.07214	0.92026	1.06585	0.13930	32.42845
DWSDiff	**0.97427**	**0.55329**	0.01635	**1.52756**	**0.95791**	**1.51121**	**0.53694**	**67.20618**

TABLE V

$AUC_{(D,F)}$ RESULTS FOR 21 HAD METHODS ACROSS THE ABU DATASET. THE TOP THREE PERFORMING METHODS ARE HIGHLIGHTED IN BOLD RED, GREEN, AND BLUE IN EACH COLUMN

Methods	Airport				Beach				Urban					Mean
	1	2	3	4	1	2	3	4	1	2	3	4	5	
GRX	0.82209	0.84038	0.92881	0.95260	0.98072	0.91058	0.99985	0.95380	0.99065	0.99463	0.95128	**0.98872**	**0.96921**	0.94487
LRX	0.91787	0.87145	0.93703	0.82288	0.96315	**0.95677**	**0.99989**	0.89671	**0.99514**	0.92510	0.97544	0.91965	0.94619	0.93287
WRX	0.85022	0.78984	0.90449	0.68205	0.94996	0.90048	0.71210	0.87715	0.98580	**0.99562**	0.37058	0.98472	0.94107	0.84185
2S-GLRT	0.88438	0.88617	0.92204	0.80791	**0.99868**	0.82659	0.99953	0.96018	**0.99697**	0.89148	0.94856	0.89962	0.92446	0.91897
FrFE	0.89658	0.96067	0.92913	0.97723	0.97551	0.91296	0.99970	0.95396	0.96309	0.99481	0.96423	0.97175	0.96031	0.95846
KIFD	**0.95159**	**0.97427**	**0.96390**	0.96675	0.98859	**0.99020**	0.99281	0.82693	0.97815	0.78740	0.97183	0.97963	**0.98531**	0.95057
LRASR	0.85729	0.84150	0.88663	0.92398	0.97370	**0.95112**	0.99968	0.96169	0.98004	0.93816	0.94986	0.92119	0.93190	0.93206
CRD	0.91131	0.94985	0.90595	0.93127	**0.99209**	0.89549	**0.99997**	0.84366	0.94502	0.91619	0.89894	0.95259	0.87143	0.92414
LSDM-MoG	0.87601	0.86872	0.88004	0.95236	0.97436	0.93539	0.99979	0.88183	0.99332	0.97302	0.97062	0.95530	0.96197	0.94021
RPCA-RX	0.81107	0.84737	0.92381	0.95362	0.98237	0.90952	0.99952	0.95906	0.99258	**0.99627**	0.95861	0.98645	0.96642	0.94513
GTVLRR	**0.95128**	**0.97834**	0.90746	**0.99698**	0.95934	0.92788	0.99240	0.97823	0.90723	0.92113	0.97824	0.88235	0.90771	0.94528
GoDec-RX	0.89881	0.93389	**0.94691**	0.96028	0.97076	0.91548	0.99938	0.97141	0.99358	0.99626	0.96811	0.98081	0.95657	**0.96094**
PTA	0.73311	0.90958	0.54759	**0.99554**	0.91841	0.82996	0.92027	0.90614	0.90547	0.97702	0.83463	0.82569	0.82578	0.85609
GAED	0.80750	0.96684	0.86246	0.96844	0.92214	0.51278	0.98935	0.93592	0.94367	0.99588	**0.98357**	**0.99255**	0.91725	0.90756
RGAE	0.63828	0.74774	0.88735	0.75452	0.93891	0.90206	0.86638	0.90419	0.98222	0.99935	0.81723	**0.99482**	0.95692	0.87615
Auto-AD	0.93685	0.89817	0.86724	0.97349	0.98453	0.92243	0.99903	**0.97908**	0.98397	0.99363	**0.98192**	0.98655	0.84674	0.95028
DFAN	0.90830	0.94546	0.94418	**0.99361**	0.99035	0.94484	**0.99993**	0.97829	0.98650	0.97682	0.97361	0.96717	0.94265	**0.96552**
DirectNet	0.89611	0.89030	0.90206	0.96389	0.97970	0.90446	0.99487	**0.98879**	0.98868	0.98855	0.97445	0.96573	0.93756	0.95193
BSDM	0.82202	0.84038	0.92882	0.95259	0.98068	0.91057	0.99985	0.95380	0.99065	0.99462	0.95129	**0.98872**	**0.96920**	0.94486
TGFA	0.84745	0.96631	0.96287	0.98637	0.72481	0.88485	0.99803	0.81898	**0.99639**	**0.99810**	0.93827	0.97093	0.95176	0.92655
DWSDiff	**0.95292**	**0.99043**	**0.97271**	0.90511	**0.99929**	0.94690	0.99938	**0.98704**	0.99360	0.99013	**0.98648**	0.98825	0.95321	**0.97427**

method consistently delivers high performance across all scenes. While some methods may achieve slightly better results in specific scenarios, DWSDiff demonstrates a more balanced and reliable performance overall. In particular, the

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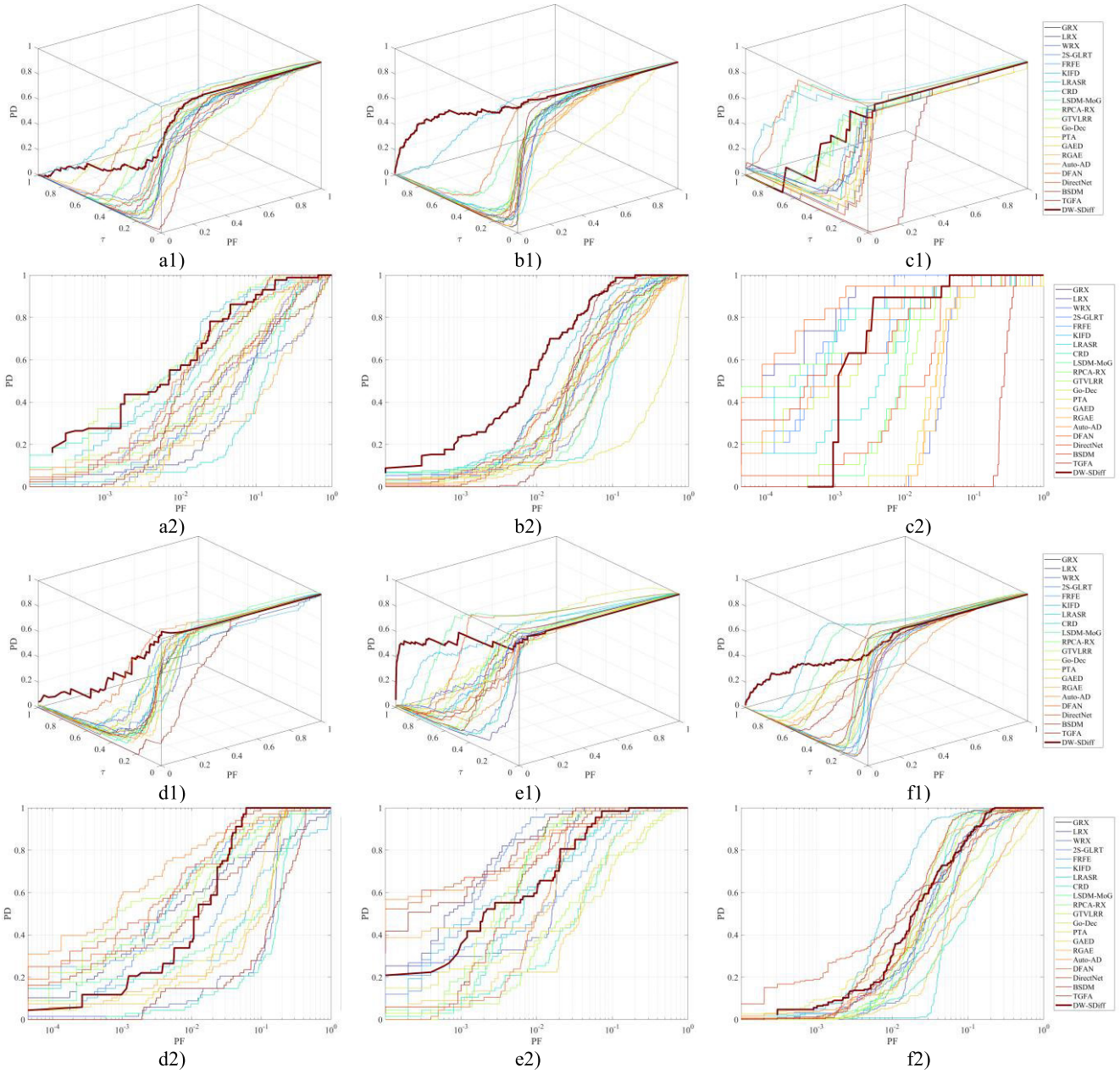


Fig. 7. 3-D ROC curve for HAD using 21 methods in (a1) Airport-1, (b1) Airport-3, (c1) Beach-1, (d1) Beach-4, (e1) Urban-1, and (f1) Urban-3. Corresponding 2-D ROC curves are shown in (a2)–(f2). The bold red line represents the ROC curve for the proposed DWSDiff in this article.

DWSDiff method consistently maintains narrow green boxes that are well-separated from the red boxes, indicating its robustness in accurately detecting anomalies while suppressing background noise. This consistent performance across various scenes highlights the DWSDiff method's adaptability and makes it a strong candidate for applications requiring reliable AD across different environments.

C. Ablation Study

In this section, we evaluate the contributions of the anomaly synthesis strategy, the diffusion model, and the loss function to the proposed method across all datasets. The results of the ablation experiments are shown in Table VI. The shaded row, corresponding to Experiment 3, represents the standard method used in this article.

1) Anomaly Synthesis Strategy: We propose an anomaly synthesis strategy based on PCA decomposition and spectral mixing models and compare it with conventional methods, such as random spectral generation or spectral dimension shuffling [65]. This corresponds to Experiments 1–3.

From the table, it is evident that the method using PCA and LSMM (E4 3) achieves the highest mAUC of 0.97427, outperforming both the random spectral generation (Experiment 1) and spectral shuffling (Experiment 2), which achieved mAUC scores of 0.95077 and 0.94831, respectively. This indicates that the proposed strategy generates data that align with the distinctive features of anomalies, allowing the network to better learn the spatial–spectral differences between anomalies and the background, resulting in more accurate detection.

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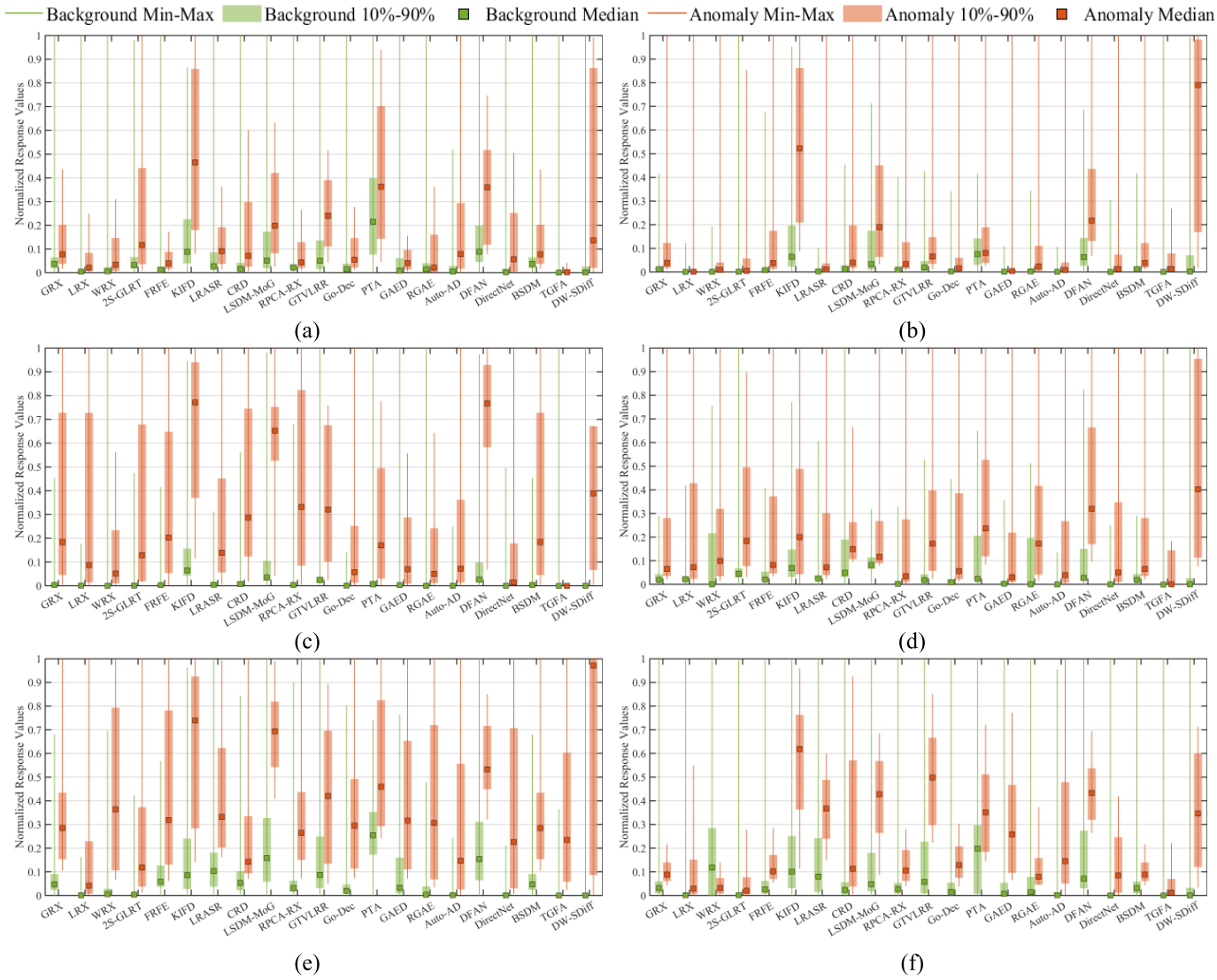


Fig. 8. Separability maps for three different scenes. (a) Airport-1. (b) Airport-3. (c) Beach-1. (d) Beach-4. (e) Urban-1. (f) Urban-3.

TABLE VI

DETECTION PERFORMANCE OF THE PROPOSED DWSDIFF HAD METHOD, SHOWCASING THE CONTRIBUTIONS OF THE ANOMALY SYNTHESIS STRATEGY, DIFFUSION MODEL, AND LOSS FUNCTION. THE SHADED ROW, CORRESPONDING TO EXPERIMENT 3, REPRESENTS THE STANDARD METHOD USED IN THIS ARTICLE

Exp. id	Anomaly Synthetic strategy			Diffusion Model			Loss Function		$mAUC_{(D,F)}$
	Random Spectral	Spectral Shuffle	PCA & LSMM	DDPM	SDiff	DWSDiff	$\mathcal{L}_{rec}$	$\mathcal{L}_{abundance}$	
1	√					√	√		0.95077
2		√				√	√		0.94831
3			√			√	√		**0.97427**
4			√						0.89345
5			√	√			√		0.95660
6			√		√		√		0.96358
7			√			√		√	0.95182

2) Diffusion Model: The proposed DWSDiff is compared against the approaches without background estimation, the DDPM, and the single-window diffusion (SDiff). This corresponds to Experiments 3–6. Notably, in Experiment 4, where background estimation is excluded and only the target detection network is trained, there is no corresponding reconstruction loss function for the abundance estimation network.

From the table, it is evident that the DWSDiff model used in Experiment 3 achieves the highest mAUC of 0.97427, significantly outperforming the DDPM (Experiment 5) with

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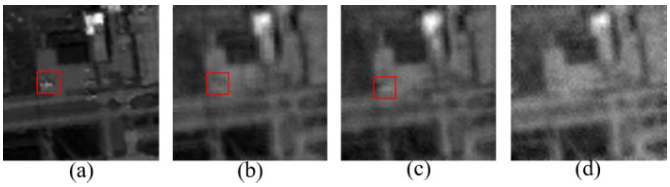


Fig. 9. Background estimation results of Airport-1 in band 10. (a) Original image. (b) DWSDiff. (c) SDiff. (d) DDPM.

an mAUC of 0.95660 and the SDiff (Experiment 6) with an mAUC of 0.96358. The comparison method without background estimation using only the detection network (Experiment 4) performs the worst with an mAUC of 0.89345. Fig. 9 shows the background estimation results of Airport-1 in band 10. The DDPM model has lower background estimation accuracy. DWSDiff achieves the most precise results. Although SDiff performs well in detection, it is still influenced by the expanded regions of anomalies within the neighborhood.

These experimental results align with the underlying characteristics of our diffusion models in terms of spectral and spatial feature processing. From (20), during the diffusion process, the image in the spatial dimension gradually transforms into random noise. In the spectral dimension, the spectral features of individual pixels are gradually overshadowed by those of their neighboring pixels over multiple diffusion steps. As a result, during the reverse reconstruction process, the model tends to reconstruct pure background with accurate spectral information, rather than the anomalies. This diffusion approach is more effective than DDPM methods in suppressing anomalies and preserving the background's spectral information.

3) Loss Function: The loss function combines reconstruction loss and detection loss, with the reconstruction loss playing a crucial role in accurately estimating the background. According to the principles of the reconstruction-based HAD method, the training phase uses MSE-based reconstruction loss $\mathcal{L}_{\text{rec}}$ between the reconstructed image and the corresponding anomaly-free image in our method. Similar to DDPM, the loss function $\mathcal{L}_{\text{abundance}}$ can be employed, calculating MSE between the abundance matrix $\mathbf{A}$ and the estimated abundance matrix predicted by $\mathcal{U}_a$. This corresponds to Experiments 3 and 7.

From the table, it is evident that $\mathcal{L}_{\text{rec}}$ used in Experiment 3 yields a higher mAUC of 0.97427, compared to Experiment 7, which uses $\mathcal{L}_{\text{abundance}}$ and achieves an mAUC of 0.95182. This indicates that directly optimizing the accuracy of background estimation leads to better AD performance than optimizing the accuracy of abundance matrix estimation. This is because the estimation of the abundance matrix does not utilize information from anomaly-free data, leading the estimated background to more closely resemble the image with synthesized anomalies.

D. Algorithm Complexity and Inference Time Analysis

The computational complexity of DWSDiff consists of two main components: the diffusion process and the reconstruction step. The overall complexity primarily depends on the window size ($T = R^2 - r^2$), the spectral dimension (L), and the spatial dimensions of the HSI ($M \times N$).

In the diffusion update step, according to $\mathbf{x}_t = \mathbf{x}_0 + \sqrt{\bar{\boldsymbol{\beta}}_t}a$, where $\beta_i = \mathbf{w}_i^2$, each pixel requires computing the cumulative weight $\bar{\boldsymbol{\beta}}_t$ over the window. The per-pixel computation involves $O(L \times T)$ operations. The total complexity of the diffusion update for the entire image is $O(M \times N \times (T \times L))$. Thus, for an HSI of size $M \times N \times L$ and $R^2 - r^2$ diffusion steps, the total complexity is $O((R^2 - r^2) \times M \times N \times L)$.

In the reconstruction step, each pixel is processed using $\mathbf{x}_{t-1} = \mathbf{x}_t - \boldsymbol{\beta}_t / \sqrt{\bar{\boldsymbol{\beta}}_t} \cdot \mathcal{U}_a(\mathbf{x}_t, t)$. The elementwise operations $\beta_t / \sqrt{\bar{\boldsymbol{\beta}}_t}$ and the subtraction operation have a complexity of $O(L)$ per pixel. As the abundance estimation network $\mathcal{U}_a$ is implemented using a traditional U-Net, its complexity is not included. For the entire image and $R^2 - r^2$ reconstruction steps, the total complexity is $O((R^2 - r^2) \times M \times N \times L)$.

The final detection result is obtained using the detection network $\mathcal{U}_M$, which is also implemented using a traditional U-Net and is, therefore, excluded from the analysis. Since the diffusion and reconstruction steps are executed sequentially, the overall complexity of the inference process is $O((R^2 - r^2) \times M \times N \times L)$.

To further assess the efficiency of our proposed method, we compare its inference time with other DL-based AD algorithms that also utilize background reconstruction strategies. As shown in Table VII, the first three columns present the inference time for Airport1, Beach1, and Urban1, while the last column shows the average inference time across all 13 images in the ABU dataset. While our method is not the fastest, its performance remains highly competitive and noteworthy.

TABLE VII
INFERENCE TIME OF DIFFERENT ALGORITHMS ON THE ABU DATASET

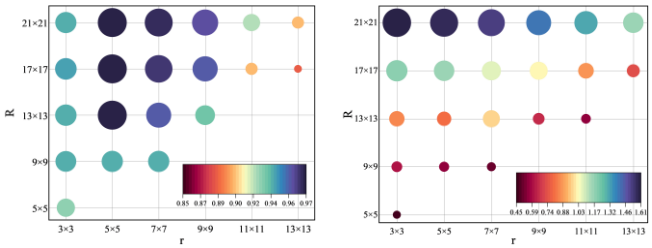
Method	Airport1	Beach1	Urban1	Mean Time(s)
GAED	0.0300	0.0496	0.0257	0.0289
RGAE	0.0135	0.0251	0.0137	0.0141
Auto-AD	**0.0020**	**0.0032**	**0.0020**	**0.0022**
DFAN	0.0232	0.0448	0.0269	0.0264
DirectNet	0.0145	0.02839	0.0153	0.0145
BSDM	1.2306	1.2740	1.2509	1.2421
TGFA	1.6190	1.5505	1.5236	1.5392
DWSDiff	0.8016	0.8169	0.8192	0.8094

E. Parameter Sensitivity Analysis

In this section, we conduct a comprehensive analysis to investigate the effects of the inner window width r and the outer window width R on the performance and computational efficiency of our proposed method.

The inner window plays a critical role in excluding target pixels during the diffusion process, thereby mitigating the contamination of the background model by target information. However, the choice of r involves a tradeoff. If r is too small, it may fail to mask the target entirely, leading to background contamination. If r is too large, it removes a substantial

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(a) (b)

Fig. 10. Impact of window width on the proposed method. (a) $AUC_{(D,F)}$, representing detection performance. (b) Inference time, indicating computational efficiency.

portion of valid background pixels, decreasing reconstruction accuracy.

Similarly, the outer window determines the region used for background reconstruction. If R is too small, anomalies might occupy a significant portion of the window, contaminating the reconstructed background. Increasing R helps dilute the influence of target pixels within the window, enhancing background approximation. However, it is important to note that increasing R also increases the computational cost of the method.

Based on empirical observations, we set the default configuration to $r = 5$ and $R = 13$, as most anomalies in the ABU dataset are smaller than 5×5. To further evaluate the sensitivity of these parameters, we tested configurations with smaller and larger window sizes. The results are visualized in Fig. 10 using matrix bubble plots. Larger circles with darker blue colors indicate a higher number. Larger circles with darker blue colors indicate larger numerical values, while smaller circles with lighter colors, trending toward red, represent smaller numerical values. From the plots, it is evident that $r = 5$ and $R = 13$ achieve a good balance between detection performance and computational efficiency.

VI. Conclusion

In this article, we introduced the DWSDiff-HAD, a novel approach designed to address the challenges of accurate background estimation in HAD. Our method integrates a spectral diffusion model with a dual-window strategy, effectively enhancing background suppression and improving AD accuracy. In addition, we developed an anomaly generation strategy based on PCA and the LSMM to create realistic training data, overcoming the scarcity of labeled hyperspectral data.

The DWSDiff-HAD method was extensively evaluated on the ABU hyperspectral datasets, where it demonstrated superior performance compared to 20 SOTA HAD methods across multiple metrics. The results highlight the robustness and effectiveness of our approach in detecting small, sparse anomalies in complex hyperspectral scenes.

Future work will focus on addressing the potential loss of spatial–spectral information caused by resampling images to specific dimensions. We aim to develop a more flexible network framework that can adapt to HSIs of varying dimensions, thereby enhancing the effectiveness and flexibility of AD tasks.

Appendix

Given the length and complexity of the derivation of (15), the detailed steps have been provided here.

To simplify the notation, all vector operations in the equations are elementwise. The derivation process for transitioning from $\mathbf{x}_0$ and $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$, using Bayes' theorem, can be expressed as follows:

$$
\begin{aligned}
&q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \\
&= \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\
&= \frac{\mathcal{N}(\mathbf{x}_t; \mathbf{x}_{t-1}, \boldsymbol{\beta}_t)\mathcal{N}(\mathbf{x}_{t-1}; \mathbf{x}_0, \bar{\boldsymbol{\beta}}_{t-1})}{\mathcal{N}(\mathbf{x}_t; \mathbf{x}_0, \bar{\boldsymbol{\beta}}_t)} \\
&= \exp\left\{-\left[\frac{(\mathbf{x}_t - \mathbf{x}_{t-1})^2}{2\sigma^2\boldsymbol{\beta}_t} + \frac{(\mathbf{x}_{t-1} - \mathbf{x}_0)^2}{2\sigma^2\bar{\boldsymbol{\beta}}_{t-1}} - \frac{(\mathbf{x}_t - \mathbf{x}_0)^2}{2\sigma^2\bar{\boldsymbol{\beta}}_t}\right]\right\} \\
&= \exp\left\{-\frac{1}{2\sigma^2}\left[\frac{\mathbf{x}_t^2 - 2\mathbf{x}_t\mathbf{x}_{t-1} + \mathbf{x}_{t-1}^2}{\boldsymbol{\beta}_t} + \frac{\mathbf{x}_{t-1}^2 - 2\mathbf{x}_0\mathbf{x}_{t-1} + \mathbf{x}_0^2}{\bar{\boldsymbol{\beta}}_{t-1}} \right.\right. \\
&\qquad \left.\left. - \frac{\mathbf{x}_t^2 - 2\mathbf{x}_0\mathbf{x}_t + \mathbf{x}_0^2}{\bar{\boldsymbol{\beta}}_t}\right]\right\} \\
&= \exp\left\{-\frac{1}{2\sigma^2}\left[\frac{-2\mathbf{x}_t\mathbf{x}_{t-1} + \mathbf{x}_{t-1}^2}{\boldsymbol{\beta}_t} + \frac{\mathbf{x}_{t-1}^2 - 2\mathbf{x}_0\mathbf{x}_{t-1}}{\bar{\boldsymbol{\beta}}_{t-1}} \right.\right. \\
&\qquad \left.\left. + C(\mathbf{x}_0, \mathbf{x}_t)\right]\right\} \\
&\propto \exp\left\{-\frac{1}{2\sigma^2}\left[-\frac{2\mathbf{x}_t\mathbf{x}_{t-1}}{\boldsymbol{\beta}_t} + \frac{\mathbf{x}_{t-1}^2}{\boldsymbol{\beta}_t} + \frac{\mathbf{x}_{t-1}^2}{\bar{\boldsymbol{\beta}}_{t-1}} - \frac{2\mathbf{x}_0\mathbf{x}_{t-1}}{\bar{\boldsymbol{\beta}}_{t-1}}\right]\right\} \\
&= \exp\left\{-\frac{1}{2\sigma^2}\left[\left(\frac{1}{\boldsymbol{\beta}_t} + \frac{1}{\bar{\boldsymbol{\beta}}_{t-1}}\right)x_{t-1}^2 - 2\left(\frac{\mathbf{x}_t}{\boldsymbol{\beta}_t} - \frac{\mathbf{x}_0}{\bar{\boldsymbol{\beta}}_{t-1}}\right)\mathbf{x}_{t-1}\right]\right\} \\
&= \exp\left\{-\frac{1}{2\sigma^2}\left[\frac{\bar{\boldsymbol{\beta}}_t}{\boldsymbol{\beta}_t\bar{\boldsymbol{\beta}}_{t-1}}\mathbf{x}_{t-1}^2 - 2\left(\frac{\mathbf{x}_t}{\boldsymbol{\beta}_t} + \frac{\mathbf{x}_0}{\bar{\boldsymbol{\beta}}_{t-1}}\right)\mathbf{x}_{t-1}\right]\right\} \\
&= \exp\left\{-\frac{1}{\frac{2\sigma^2\boldsymbol{\beta}_t\bar{\boldsymbol{\beta}}_{t-1}}{\bar{\boldsymbol{\beta}}_t}}\left[\mathbf{x}_{t-1}^2 - 2\frac{\bar{\boldsymbol{\beta}}_{t-1}x_t + \boldsymbol{\beta}_t\mathbf{x}_0}{\bar{\boldsymbol{\beta}}_t}\mathbf{x}_{t-1}\right]\right\} \\
&\qquad \sim \mathcal{N}\left(\mathbf{x}_{t-1}; \frac{\bar{\boldsymbol{\beta}}_{t-1}\mathbf{x}_t + \boldsymbol{\beta}_t\mathbf{x}_0}{\bar{\boldsymbol{\beta}}_t}, \frac{\sigma^2\boldsymbol{\beta}_t\bar{\boldsymbol{\beta}}_{t-1}}{\boldsymbol{\beta}_t}\right)
\end{aligned} \tag{34}
$$

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